UROS-26 Investigating Comparative Performance of Numerical Integrators on a Hamiltonian Two-Body System Under Newtonian Mechanics

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1 ABSTRACT

The celestial dynamics of the N-body problem are most commonly studied through a restricted set of numerical integration methods, often tracking only a limited range of physical variables. Existing literature focuses predominantly on symplectic integrators, including Leapfrog, Velocity Verlet, Symplectic Euler, and the fourth-order Runge-Kutta method. This paper presents a comparative evaluation of five numerical integration schemes: Explicit Euler, Symplectic Euler, Velocity Verlet, Leapfrog, and Runge-Kutta 4th Order with respect to energy conservation, linear and angular momentum conservation, and computational efficiency within a Hamiltonian framework. Building on this comparison, we further investigate a modification of the best-performing integrator, targeting improvements in energy, angular momentum, and orbital stability for the two-body problem. The trade-offs between numerical precision and computational cost introduced by this modification are examined in the context of a high-performance implementation. Our findings have broader implications for the simulation of celestial mechanics across a range of cosmological settings, with particular emphasis on two-body classical gravitational dynamics.

2 INTRODUCTION

When studying N-body celestial systems, selecting an appropriate numerical integration scheme is fundamental to obtaining accurate and meaningful results. Leapfrog and fourth-order Runge-Kutta (RK4) are routinely employed as standard integrators across modern computational astrophysics, particularly in non-chaotic regimes.

This paper investigates the two-body gravitational problem within the frameworks of Newtonian mechanics and Hamiltonian formalism. In classical mechanics, the Hamiltonian is a function describing the total energy of a system in terms of its generalised coordinates and conjugate momenta, governing the time evolution of the system through Hamilton's equations. This formalism is particularly well-suited to the study of orbital dynamics, as it naturally exposes conserved quantities such as energy and angular momentum against which numerical schemes can be rigorously evaluated.

We restrict our analysis to classical mechanics as defined by Newton and Kepler, applying the governing equations of the field accordingly. A high-performance simulation engine is developed in C++20 to carry out procedurally generated two-body simulations, with the engine exposed to Python 3.12 via a foreign function interface – a common and practical convention in scientific computing workflows. To make the resulting dual-resolution study tractable, simulations are distributed across a fixed-size worker pool, with each worker independently generating a problem instance and executing all five candidate integrators under matching initial conditions. Simulation data is then analysed in Python to draw conclusions regarding integrator accuracy and computational efficiency.

Having identified the most suitable standard integrator, we subsequently construct a higher-order scheme and repeat the analysis under equivalent conditions. Final conclusions are drawn on the basis of energy and angular momentum conservation behaviour, alongside a comparison of computational cost across all tested methods.

3 ANALYTICAL SOLUTIONS

3.1 NORMALISED SYSTEM

Aligning with the goals of this research we will show the analytical solutions for a singular problem setup across five integration schemes under ten timesteps.

Before starting any calculations, it is required to apply the appropriate units of measurement. It was decided to apply a normalised system due to the ease of debugging and interpretation, numerical stability and avoiding the vulnerability of floating-point errors.

Hence why, the units are as follows:

- Distance – Astronomical Units (Earth to Sun Distance)
- Mass – Solar Masses
- Gravitational Attraction – $4\pi^2$
- Timestep – Years

Following studies a timestep of $\Delta t = 10^{-3}\,\mathrm{yr}$ has been selected, since unstable integrators tend to converge later in the simulation producing more reasonable data. This allows tracking of convergences and energy preservation more accurately.

Due to the assumption of a normalised numerical system the value of Gravitational Attraction Constant (G) has to be derived from those systemic values. Physics remains the same, while the measuring system modifies. Since the normalised system is based on the relationship between the Earth and the Sun, we can derive the value of G that complies with the numerical system.

Applying Newtonian gravity,

$$F = G\frac{m_1 m_2}{r^2}, \tag{1}$$

and the equation for circular motion,

$$F = \frac{m_2 v^2}{r}, \tag{2}$$

we obtain

$$\frac{m_2 v^2}{r} = G\frac{m_1 m_2}{r^2}. \tag{3}$$

Cancelling m_2 from both sides yields

$$v^2 = \frac{G m_1}{r}. \tag{4}$$

Since the circumference of a circular orbit is $2\pi r$, the orbital velocity can be expressed in terms of the orbital period T as

$$v = \frac{2\pi r}{T}. \tag{5}$$

Substituting this expression for v into Equation (3) gives

$$\left(\frac{2\pi r}{T}\right)^2 = \frac{G m_1}{r}. \tag{6}$$

Solving for T yields

$$T^2 = \frac{4\pi^2 r^3}{G m_1}. \tag{7}$$

Assuming a normalised system of units in which the orbital period, orbital radius, and central mass are all equal to unity ($T = r = m_1 = 1$), we obtain

$$1 = \frac{4\pi^2}{G}. \tag{8}$$

Rearranging gives

$$G = 4\pi^2. \tag{9}$$

3.2 ANALYTICAL PROBLEM

The problem definition is as follows: A star of one solar mass $M = 1\,M_\odot$ is fixed at the origin. A planet orbiting the star is 333000^{-1} of solar mass and is placed at initial position $\vec{r}_0 = (2,\,1)^T$ AU. All quantities are expressed in natural astronomical units: distances in AU, time in years, and mass in solar masses, with $G = 4\pi^2\,\mathrm{AU}^3\,\mathrm{yr}^{-2}\,M_\odot^{-1}$. The simulation procedurally generates two objects containing their mass and coordinates, while the rest of the calculations are presented through the later simulation.

3.3 INITIAL VALUES

Initial distance

The initial separation between the planet and the star is

$$r_0 = |\vec{r}_0| = \sqrt{x_0^2 + y_0^2} = \sqrt{2^2 + 1^2} = \sqrt{5} \approx 2.23606798\,\mathrm{AU}. \tag{10}$$

Initial acceleration

Newton's law of gravity gives the acceleration per unit planet mass as

$$\vec{a} = -\frac{GM}{r^3}\vec{r}. \tag{11}$$

Substituting $G = 4\pi^2$, $M = 1$, $r_0 = \sqrt{5}$, $\vec{r}_0 = (2, 1)^T$:

$$\vec{a}_0 = -\frac{4\pi^2}{(\sqrt{5})^3}\begin{pmatrix}2\\1\end{pmatrix} = -\frac{4\pi^2}{5\sqrt{5}}\begin{pmatrix}2\\1\end{pmatrix} \approx \begin{pmatrix}-7.06211403\\-3.53105702\end{pmatrix}\,\mathrm{AU\,yr^{-2}}. \tag{12}$$

The magnitude is $|\vec{a}_0| = 4\pi^2/5 \approx 7.8957\,\mathrm{AU\,yr^{-2}}$.

Circular orbital velocity

For a circular orbit, gravitational acceleration exactly supplies centripetal acceleration:

$$v_c = \sqrt{\frac{GM}{r_0}} = \sqrt{\frac{4\pi^2}{\sqrt{5}}} \approx 4.20181926\,\mathrm{AU\,yr^{-1}}. \tag{13}$$

The velocity is directed tangentially (perpendicular to $\vec{r}_0$, counter-clockwise):

$$\hat{t} = \frac{(-y_0,\,x_0)^T}{r_0} = \frac{1}{\sqrt{5}}\begin{pmatrix}-1\\2\end{pmatrix}, \qquad \vec{v}_0 = v_c\,\hat{t} \approx \begin{pmatrix}-1.87911070\\3.75822140\end{pmatrix}\,\mathrm{AU\,yr^{-1}}. \tag{14}$$

Initial conserved quantities

The total mechanical energy (kinetic plus potential) per unit planet mass is

$$E_0 = \frac{1}{2}m|\vec{v}_0|^2 - \frac{GMm}{r_0} = -\frac{GMm}{2r_0} \approx -2.6509 \times 10^{-5}\, M_\odot\, \mathrm{AU}^2\, \mathrm{yr}^{-2}, \quad (15)$$

and the scalar z-component of angular momentum is

$$L_0 = m(x_0 v_{y,0} - y_0 v_{x,0}) \approx 2.8215 \times 10^{-5}\, M_\odot\, \mathrm{AU}^2\, \mathrm{yr}^{-1}. \quad (16)$$

3.4 FIRST TIMESTEP DERIVATION ($\Delta t = 10^{-3}$ yr)

We adopt a time step $\Delta t = 10^{-3}$ yr and derive the first update for each integrator explicitly. Relative errors in the conserved quantities are defined as $\varepsilon_E = |E_n - E_0|/|E_0|$ and $\varepsilon_L = |L_n - L_0|/|L_0|$.

Explicit Euler

Update rule: $\vec{r}_{n+1} = \vec{r}_n + \Delta t\, \vec{v}_n, \quad \vec{v}_{n+1} = \vec{v}_n + \Delta t\, \vec{a}_n$.

Step $n = 0 \to 1$:

$$\vec{r}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 10^{-3} \begin{pmatrix} -1.87911070 \\ 3.75822140 \end{pmatrix} = \begin{pmatrix} 1.99812089 \\ 1.00375822 \end{pmatrix} \mathrm{AU},$$
$$\vec{v}_1 = \begin{pmatrix} -1.87911070 \\ 3.75822140 \end{pmatrix} + 10^{-3} \begin{pmatrix} -7.06211403 \\ -3.53105702 \end{pmatrix} = \begin{pmatrix} -1.88617281 \\ 3.75469034 \end{pmatrix} \mathrm{AU\,yr}^{-1}.$$

Both position and velocity are updated using values from the same timestep, so this method is first-order accurate and does not conserve energy over time.

Symplectic Euler

Update rule: velocity is updated first, then position uses the updated velocity: $\vec{v}_{n+1} = \vec{v}_n + \Delta t\, \vec{a}_n, \quad \vec{r}_{n+1} = \vec{r}_n + \Delta t\, \vec{v}_{n+1}$.

Step $n = 0 \to 1$:

$$\vec{v}_1 = \begin{pmatrix} -1.87911070 \\ 3.75822140 \end{pmatrix} + 10^{-3} \begin{pmatrix} -7.06211403 \\ -3.53105702 \end{pmatrix} = \begin{pmatrix} -1.88617281 \\ 3.75469034 \end{pmatrix} \mathrm{AU\,yr}^{-1},$$
$$\vec{r}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 10^{-3} \begin{pmatrix} -1.88617281 \\ 3.75469034 \end{pmatrix} = \begin{pmatrix} 1.99811383 \\ 1.00375469 \end{pmatrix} \mathrm{AU}.$$

Symplectic Euler is also first-order, but it preserves the symplectic structure of Hamiltonian mechanics, resulting in bounded (rather than drifting) energy errors.

Velocity Verlet

Update rule: $\vec{r}_{n+1} = \vec{r}_n + \Delta t\, \vec{v}_n + \frac{1}{2}\Delta t^2\, \vec{a}_n, \quad \vec{v}_{n+1} = \vec{v}_n + \frac{\Delta t}{2}(\vec{a}_n + \vec{a}_{n+1})$.

Step $n = 0 \to 1$:

$$\vec{r}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 10^{-3} \begin{pmatrix} -1.87911070 \\ 3.75822140 \end{pmatrix} + \frac{10^{-6}}{2} \begin{pmatrix} -7.06211403 \\ -3.53105702 \end{pmatrix} = \begin{pmatrix} 1.99811736 \\ 1.00375646 \end{pmatrix} \text{ AU}.$$

Then $\vec{a}_1$ is recomputed from $\vec{r}_1$, and

$$\vec{v}_1 = \begin{pmatrix} -1.87911070 \\ 3.75822140 \end{pmatrix} + \frac{10^{-3}}{2} \left[\begin{pmatrix} -7.06211403 \\ -3.53105702 \end{pmatrix} + \vec{a}_1 \right] = \begin{pmatrix} -1.88616949 \\ 3.75468371 \end{pmatrix} \text{ AU yr}^{-1}.$$

Leapfrog (Kick–Drift–Kick)

The Leapfrog integrator stores velocities at half-integer steps. The initialisation half-kick is:

$$\vec{v}_{1/2} = \vec{v}_0 + \frac{\Delta t}{2}\vec{a}_0 = \begin{pmatrix} -1.87911070 \\ 3.75822140 \end{pmatrix} + 5\times 10^{-4} \begin{pmatrix} -7.06211403 \\ -3.53105702 \end{pmatrix} = \begin{pmatrix} -1.88264677 \\ 3.75645587 \end{pmatrix} \text{ AU yr}^{-1}.$$

Full drift step $n = 0 \to 1$:

$$\vec{r}_1 = \vec{r}_0 + \Delta t\, \vec{v}_{1/2} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 10^{-3} \begin{pmatrix} -1.88264677 \\ 3.75645587 \end{pmatrix} = \begin{pmatrix} 1.99811736 \\ 1.00375646 \end{pmatrix} \text{ AU}.$$

Velocities at integer steps are recovered as $\vec{v}_n = \vec{v}_{n+1/2} - \frac{\Delta t}{2}\vec{a}_n$, producing results identical to Velocity Verlet to floating-point precision.

RK4 (4th-order Runge–Kutta)

Writing the state vector as $\mathbf{y} = (\vec{r}, \vec{v})^T$ with $\dot{\mathbf{y}} = (\vec{v}, \vec{a}(\vec{r}))^T$, four intermediate slopes are:

$$\mathbf{k}_1 = f(\mathbf{y}_n), \quad \mathbf{k}_2 = f\left(\mathbf{y}_n + \tfrac{\Delta t}{2}\mathbf{k}_1\right), \quad \mathbf{k}_3 = f\left(\mathbf{y}_n + \tfrac{\Delta t}{2}\mathbf{k}_2\right), \quad \mathbf{k}_4 = f(\mathbf{y}_n + \Delta t\, \mathbf{k}_3),$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \frac{\Delta t}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)\,.$$

After step $n = 0 \to 1$:

$$\vec{r}_1 \approx \begin{pmatrix} 1.99811736 \\ 1.00375645 \end{pmatrix} \text{ AU}, \qquad \vec{v}_1 \approx \begin{pmatrix} -1.88616949 \\ 3.75468371 \end{pmatrix} \text{ AU yr}^{-1}.$$

RK4 is fourth-order accurate; energy errors at each step are $\mathcal{O}(\Delta t^5)$.

3.5 10-STEP SIMULATION TABLES ($\Delta t = 10^{-3}$ yr)

The conserved quantities and their relative errors are defined as

$$E_n = \frac{1}{2}m\big(v_{x,n}^2 + v_{y,n}^2\big) - \frac{GMm}{r_n}, \quad L_n = m(x_n v_{y,n} - y_n v_{x,n}), \quad \varepsilon_E = \frac{|E_n - E_0|}{|E_0|}, \quad \varepsilon_L = \frac{|L_n - L_0|}{|L_0|}. \tag{17}$$

Units: positions and distances in AU, velocities in AU yr^{-1}, accelerations in AU yr^{-2}, energies in $M_\odot$ AU^2 yr^{-2}, angular momenta in $M_\odot$ AU^2 yr^{-1}.

Explicit Euler

n	$\vec{r}$ (AU)	$\vec{v}$ (AU yr^{-1})	$\vec{a}$ (AU yr^{-2})	$\lvert\vec{v}\rvert$	r	ε_E	ε_L
0	(2.00, 1.00)	(−1.88, 3.76)	(−7.06, −3.53)	4.20	2.24	0	0
1	(2.00, 1.00)	(−1.89, 3.75)	(−7.06, −3.54)	4.20	2.24	7.06×10^{-6}	3.53×10^{-6}
2	(2.00, 1.01)	(−1.89, 3.75)	(−7.05, −3.56)	4.20	2.24	1.41×10^{-5}	7.06×10^{-6}
3	(1.99, 1.01)	(−1.90, 3.75)	(−7.04, −3.57)	4.20	2.24	2.12×10^{-5}	1.06×10^{-5}
4	(1.99, 1.02)	(−1.91, 3.74)	(−7.04, −3.58)	4.20	2.24	2.82×10^{-5}	1.41×10^{-5}
5	(1.99, 1.02)	(−1.91, 3.74)	(−7.03, −3.60)	4.20	2.24	3.53×10^{-5}	1.77×10^{-5}
6	(1.99, 1.02)	(−1.92, 3.74)	(−7.02, −3.61)	4.20	2.24	4.24×10^{-5}	2.12×10^{-5}
7	(1.99, 1.03)	(−1.93, 3.73)	(−7.01, −3.62)	4.20	2.24	4.94×10^{-5}	2.47×10^{-5}
8	(1.98, 1.03)	(−1.94, 3.73)	(−7.01, −3.64)	4.20	2.24	5.65×10^{-5}	2.82×10^{-5}
9	(1.98, 1.03)	(−1.94, 3.73)	(−7.00, −3.65)	4.20	2.24	6.36×10^{-5}	3.18×10^{-5}
10	(1.98, 1.04)	(−1.95, 3.72)	(−6.99, −3.66)	4.20	2.24	7.06×10^{-5}	3.53×10^{-5}

Table 1: Explicit Euler, $\Delta t = 10^{-3}$ yr. Energy and angular momentum drift monotonically at $\mathcal{O}(\Delta t)$ per step – characteristic of a non-symplectic first-order method.

Symplectic Euler

n	$\vec{r}$ (AU)	$\vec{v}$ (AU yr^{-1})	$\vec{a}$ (AU yr^{-2})	$\lvert\vec{v}\rvert$	r	ε_E	ε_L
0	(2.00, 1.00)	(−1.88, 3.76)	(−7.06, −3.53)	4.20	2.24	0	0
1	(2.00, 1.00)	(−1.89, 3.75)	(−7.06, −3.54)	4.20	2.24	3.10×10^{-12}	~0
2	(2.00, 1.01)	(−1.89, 3.75)	(−7.05, −3.56)	4.20	2.24	1.30×10^{-11}	~0
3	(1.99, 1.01)	(−1.90, 3.75)	(−7.04, −3.57)	4.20	2.24	2.80×10^{-11}	~0
4	(1.99, 1.01)	(−1.91, 3.74)	(−7.04, −3.58)	4.20	2.24	5.00×10^{-11}	~0
5	(1.99, 1.02)	(−1.91, 3.74)	(−7.03, −3.60)	4.20	2.24	7.80×10^{-11}	~0
6	(1.99, 1.02)	(−1.92, 3.74)	(−7.02, −3.61)	4.20	2.24	1.10×10^{-10}	~0
7	(1.99, 1.03)	(−1.93, 3.73)	(−7.02, −3.62)	4.20	2.24	1.50×10^{-10}	~0
8	(1.98, 1.03)	(−1.94, 3.73)	(−7.01, −3.64)	4.20	2.24	2.00×10^{-10}	~0
9	(1.98, 1.03)	(−1.94, 3.73)	(−7.00, −3.65)	4.20	2.24	2.50×10^{-10}	~0
10	(1.98, 1.04)	(−1.95, 3.72)	(−6.99, −3.66)	4.20	2.24	3.10×10^{-10}	~0

Table 2: Symplectic Euler, $\Delta t = 10^{-3}$ yr. Angular momentum is conserved to machine precision; energy error is bounded ($\sim 10^{-10}$–10^{-12}) with no secular drift – hallmark of a symplectic method.

Velocity Verlet

n	$\vec{r}$ (AU)	$\vec{v}$ (AU yr^{-1})	$\vec{a}$ (AU yr^{-2})	$\lvert\vec{v}\rvert$	r	ε_E	ε_L
0	(2.00, 1.00)	(−1.88, 3.76)	(−7.06, −3.53)	4.20	2.24	0	0
1	(2.00, 1.00)	(−1.89, 3.75)	(−7.06, −3.54)	4.20	2.24	$\sim 10^{-16}$	~ 0
2	(2.00, 1.01)	(−1.89, 3.75)	(−7.05, −3.56)	4.20	2.24	$\sim 10^{-16}$	~ 0
3	(1.99, 1.01)	(−1.90, 3.75)	(−7.04, −3.57)	4.20	2.24	$\sim 10^{-16}$	~ 0
4	(1.99, 1.02)	(−1.91, 3.74)	(−7.04, −3.58)	4.20	2.24	$\sim 10^{-16}$	~ 0
5	(1.99, 1.02)	(−1.91, 3.74)	(−7.03, −3.60)	4.20	2.24	$\sim 10^{-16}$	~ 0
6	(1.99, 1.02)	(−1.92, 3.74)	(−7.02, −3.61)	4.20	2.24	$\sim 10^{-16}$	~ 0
7	(1.99, 1.03)	(−1.93, 3.73)	(−7.02, −3.62)	4.20	2.24	$\sim 10^{-16}$	~ 0
8	(1.98, 1.03)	(−1.94, 3.73)	(−7.01, −3.64)	4.20	2.24	$\sim 10^{-16}$	~ 0
9	(1.98, 1.03)	(−1.94, 3.73)	(−7.00, −3.65)	4.20	2.24	$\sim 10^{-16}$	~ 0
10	(1.98, 1.04)	(−1.95, 3.72)	(−6.99, −3.66)	4.20	2.24	1.00×10^{-15}	~ 0

Table 3: Velocity Verlet, $\Delta t = 10^{-3}$ yr. Both $|\vec{v}|$ and r remain constant to 3 significant figures throughout; energy errors are at floating-point rounding level ($\sim 10^{-15}$–10^{-16}). This is a second-order symplectic method.

Leapfrog (Kick–Drift–Kick)

n	$\vec{r}$ (AU)	$\vec{v}$ (AU yr^{-1})	$\vec{a}$ (AU yr^{-2})	$\lvert\vec{v}\rvert$	r	ε_E	ε_L
0	(2.00, 1.00)	(−1.88, 3.76)	(−7.06, −3.53)	4.20	2.24	0	0
1	(2.00, 1.00)	(−1.89, 3.75)	(−7.06, −3.54)	4.20	2.24	~ 0	~ 0
2	(2.00, 1.01)	(−1.89, 3.75)	(−7.05, −3.56)	4.20	2.24	$\sim 10^{-16}$	~ 0
3	(1.99, 1.01)	(−1.90, 3.75)	(−7.04, −3.57)	4.20	2.24	$\sim 10^{-16}$	~ 0
4	(1.99, 1.02)	(−1.91, 3.74)	(−7.04, −3.58)	4.20	2.24	$\sim 10^{-16}$	~ 0
5	(1.99, 1.02)	(−1.91, 3.74)	(−7.03, −3.60)	4.20	2.24	$\sim 10^{-16}$	~ 0
6	(1.99, 1.02)	(−1.92, 3.74)	(−7.02, −3.61)	4.20	2.24	$\sim 10^{-16}$	~ 0
7	(1.99, 1.03)	(−1.93, 3.73)	(−7.02, −3.62)	4.20	2.24	$\sim 10^{-16}$	~ 0
8	(1.98, 1.03)	(−1.94, 3.73)	(−7.01, −3.64)	4.20	2.24	$\sim 10^{-16}$	~ 0
9	(1.98, 1.03)	(−1.94, 3.73)	(−7.00, −3.65)	4.20	2.24	$\sim 10^{-16}$	~ 0
10	(1.98, 1.04)	(−1.95, 3.72)	(−6.99, −3.66)	4.20	2.24	$\sim 10^{-16}$	~ 0

Table 4: Leapfrog (KDK), $\Delta t = 10^{-3}$ yr. Results are numerically identical to Velocity Verlet (they are algebraically equivalent for constant-mass conservative systems). The Leapfrog is distinguished by its staggered storage of velocities at half-integer steps.

RK4 (4th-order Runge–Kutta)

n	$\vec{r}$ (AU)	$\vec{v}$ (AU yr^{-1})	$\vec{a}$ (AU yr^{-2})	$\lvert\vec{v}\rvert$	r	ε_E	ε_L
0	(2.00, 1.00)	(−1.88, 3.76)	(−7.06, −3.53)	4.20	2.24	0	0
1	(2.00, 1.00)	(−1.89, 3.75)	(−7.06, −3.54)	4.20	2.24	~ 0	~ 0
2	(2.00, 1.01)	(−1.89, 3.75)	(−7.05, −3.56)	4.20	2.24	~ 0	~ 0
3	(1.99, 1.01)	(−1.90, 3.75)	(−7.04, −3.57)	4.20	2.24	~ 0	~ 0
4	(1.99, 1.02)	(−1.91, 3.74)	(−7.04, −3.58)	4.20	2.24	~ 0	~ 0
5	(1.99, 1.02)	(−1.91, 3.74)	(−7.03, −3.60)	4.20	2.24	~ 0	~ 0
6	(1.99, 1.02)	(−1.92, 3.74)	(−7.02, −3.61)	4.20	2.24	$\sim 10^{-16}$	~ 0
7	(1.99, 1.03)	(−1.93, 3.73)	(−7.02, −3.62)	4.20	2.24	$\sim 10^{-16}$	~ 0
8	(1.98, 1.03)	(−1.94, 3.73)	(−7.01, −3.64)	4.20	2.24	~ 0	~ 0
9	(1.98, 1.03)	(−1.94, 3.73)	(−7.00, −3.65)	4.20	2.24	~ 0	~ 0
10	(1.98, 1.04)	(−1.95, 3.72)	(−6.99, −3.66)	4.20	2.24	~ 0	~ 0

Table 5: RK4, $\Delta t = 10^{-3}$ yr. Both $|\vec{v}|$ and r are constant to 3 significant figures throughout; energy errors are entirely at floating-point rounding level. RK4 is not symplectic but is 4th-order accurate, so local errors are $\mathcal{O}(\Delta t^5)$.

3.6 SUMMARY OF INTEGRATOR PERFORMANCE

Integrator	Order	ε_E at $n = 10$	Conservation behaviour
Explicit Euler	1	7.06×10^{-5}	Monotonic drift (non-symplectic)
Symplectic Euler	1	3.10×10^{-10}	Bounded oscillation (symplectic)
Velocity Verlet	2	$\sim 10^{-15}$	Near machine precision (symplectic)
Leapfrog	2	$\sim 10^{-16}$	Identical to Velocity Verlet
RK4	4	$\sim 10^{-16}$	Near machine precision (not symplectic)

Table 6: Energy relative error ε_E at $n = 10$ for each integrator with $\Delta t = 10^{-3}$ yr.

Key observations:

- **Explicit Euler** loses energy monotonically at rate $\mathcal{O}(\Delta t)$ per step – unacceptable for long-term orbital integration. At the finer $\Delta t = 10^{-3}$, however, the error at $n = 10$ is 7.06×10^{-5}, two orders of magnitude smaller than at $\Delta t = 10^{-2}$.
- **Symplectic Euler** oscillates around the true energy with no secular drift, despite having the same first-order cost. Energy error ($\sim 3 \times 10^{-10}$) is orders of magnitude lower than Explicit Euler; angular momentum is conserved to machine precision.
- **Velocity Verlet** and **Leapfrog** are algebraically equivalent and provide near-machine-precision conservation at second-order cost – the standard choice in N-body simulation. At $\Delta t = 10^{-3}$ the energy error is entirely at the floating-point rounding floor ($\sim 10^{-15}$–10^{-16}).

- **RK4** gives the lowest error per step at this timestep, with energy errors indistinguishable from machine epsilon. However, RK4 is not symplectic, so secular energy drift will appear on very long timescales; it also requires four force evaluations per step (versus one for Euler and two for Verlet).

4 Methods

This investigation follows a structured methodology designed to minimise bias and maximise the accuracy and reproducibility of results. Synthetic data are generated using a custom C++ simulator across a variable number of timesteps and procedurally constructed initial problem configurations.

4.1 Procedural Generation

Initial conditions are procedurally generated using a Mersenne–Twister pseudo-random number engine with fixed seed parameters to ensure reproducibility. Two physical quantities require generation for each system: the mass of the central object and the spatial coordinates of the orbiting body.

Central body mass. The stellar mass is sampled uniformly from the range $[1.0, 2.4]\,M_{\odot}$. This interval is chosen to restrict the simulation domain to main-sequence stars of spectral classes G through A, encompassing objects analogous to the Sun and Vega, respectively. This restriction is physically motivated: the gravitational influence of more massive stars (class O or B) would cause rapid orbital decay or ejection, while sub-solar-mass stars would introduce qualitatively different dynamical regimes outside the scope of this study.

Planetary mass. The mass of each orbiting body is derived from the stellar mass via the fixed ratio

$$m_{\text{planet}} = \frac{M_{\star}}{333{,}000}, \tag{18}$$

which approximates the observed mass ratio between Earth and the Sun, providing a physically realistic planetary-mass analogue for each generated system.

Orbital coordinates. The initial separation between the central body and the orbiting object is sampled uniformly from the range $[1, 5]$ AU. This interval is deliberately chosen to prevent premature convergence in the numerical integrator and to ensure a well-conditioned, stable dataset throughout the integration period.

4.2 Simulation

Following procedural generation, the derived initial quantities velocity, acceleration, and separation distance are computed analytically, alongside baseline stability diagnostics for angular momentum and total energy.

4.2.1 Parallel Compute Pipeline

Simulations are executed concurrently via a fixed-size worker pool of four CPU cores, with each worker assigned an independently generated orbital problem subject to the physical constraints described above. Although, during the research process certain empirical findings were discovered, hence the cores have been reduced to three to supplement RAM use of the simulating device. For every assigned problem, the worker executes all five candidate integrators – Explicit Euler, Symplectic Euler, Leapfrog, Velocity Verlet, and RK4 – under identical initial conditions, ensuring that comparisons between schemes are made on a like-for-like basis.

There are 120 problems to be solved.

Resolution study. For each integrator, three solutions are computed:

- **Baseline resolution:** $N = 10^5$ steps with $\Delta t = 10^{-1}$.
- **Medium resolution:** $N = 10^6$ steps with $\Delta t = 10^{-2}$.
- **High Resolution:** $N = 10^7$ steps with $\Delta t = 10^{-3}$.

The higher-resolution solution serves as the reference trajectory for convergence analysis, against which the baseline solution is compared. As both regimes share the product $N \cdot \Delta t$, this comparison isolates the effect of step size on integrator accuracy while holding the total integrated time constant.

Data collection. For each simulation run, the raw trajectory is recorded at every timestep, comprising time, position, velocity, total energy, angular momentum, and any additional state variables required for downstream diagnostics. Each run is written to an individual `.csv` file, preserving full trajectory resolution for subsequent analysis. The summary table of each run will be produced, containing information on the ID of each simulation, angular momentum and energy conservation, and compute time for each integrator.

Graph plots of the trajectories will be produced with both timesteps, however the conservation and relative error data will be plotted from the summary table.

Summary statistics. For each simulation, the following scalar metrics are computed and appended to a master summary table: runtime, maximum and mean relative energy error, and maximum and mean relative angular momentum error. This table forms the basis for all cross-integrator comparisons reported in later sections.

Analysis and visualisation. Post-processing and visualisation are performed using `matplotlib` under Python 3.12. Three categories of plot are produced:

1. *Trajectory-level plots* for each integrator: orbital trajectory in the *xy*-plane, energy conservation over time, angular momentum conservation over time, relative energy error over time, and relative angular momentum error over time.

2. *Summary comparison plots* across integrators: average runtime, maximum and mean relative energy error, and maximum and mean relative angular momentum error.

3. *Convergence plots*: an overlay of trajectories at $\Delta t = 10^{-1}$, $\Delta t = 10^{-2}$ and $\Delta t = 10^{-3}$, a comparison of numerical drift between the two resolutions, and a comparison of accuracy versus computational cost across all integrators.

4.3 Integrator Derivation

In line with one of the primary objectives of this research the development and evaluation of novel numerical schemes, the best-performing integrator identified among the five candidates will be extended to a higher order via appropriate composition or correction methods. This derived integrator will then be benchmarked against its lower-order counterpart under identical simulation conditions.

4.4 Hypothesis

Drawing on prior analytical work and the existing literature on geometric numerical integration, the primary hypothesis is that the *Velocity Verlet* integrator will deliver the most favourable trade-off between computational efficiency and energy fidelity. This expectation is grounded in two properties of the method: its symplecticity, which guarantees long-term conservation of the symplectic structure of Hamiltonian systems, and its second-order accuracy, which provides a sufficient resolution for orbital dynamics without the overhead of higher-order schemes.

Integrator performance will be evaluated against three criteria: (i) long-term conservation of total energy, (ii) preservation of angular momentum, and (iii) computational cost, measured in wall-clock time per integration step.

5 Preliminary Results

After conducting the first experiment to identify the best performing integrator across higher-described parameters, the following was discovered.

5.1 Runtime

As shown on the Figure 1, we may observe Leapfrog being the fastest performing integrator at higher resolutions. The result is close to the Velocity Verlet integrator, although the two integrators are mathematically related. RK4 shows the highest possible runtime due to the fact of it having two more orders.

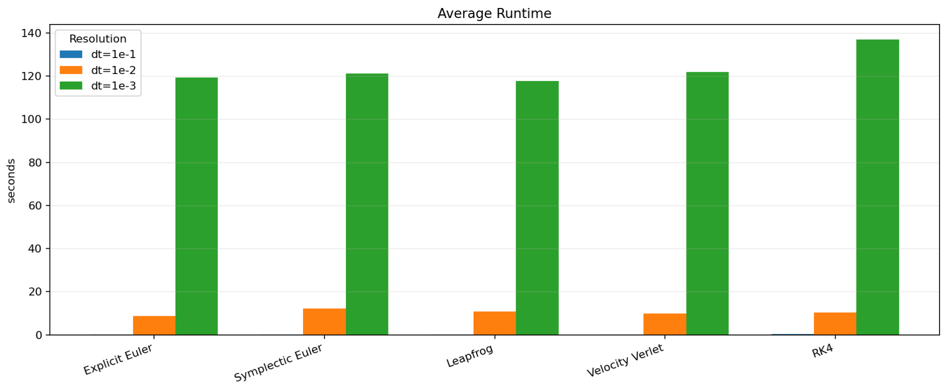

Figure 1: Average Integrator Runtime

5.2 Energy Conservation

Figure 2 presents the distribution of relative energy errors for each integrator across the three tested timesteps. Explicit Euler sits consistently around a relative error of 10^0 regardless of resolution, with a very narrow spread, indicating that the method fails to conserve energy irrespective of how finely the timestep is refined. Symplectic Euler, Leapfrog, and Velocity Verlet all show a clear downward trend in error as the timestep decreases, which is expected of lower-order methods whose local truncation error scales with Δt. RK4 displays the widest spread of the five integrators, ranging from below 10^{-12} at $\Delta t = 10^{-3}$ up to over 10^1 at $\Delta t = 10^{-1}$, showing that its performance is highly sensitive to the choice of timestep.

Figure 3 shows the mean relative energy error plotted directly against timestep on a log-log scale. Explicit Euler remains flat at approximately 10^0

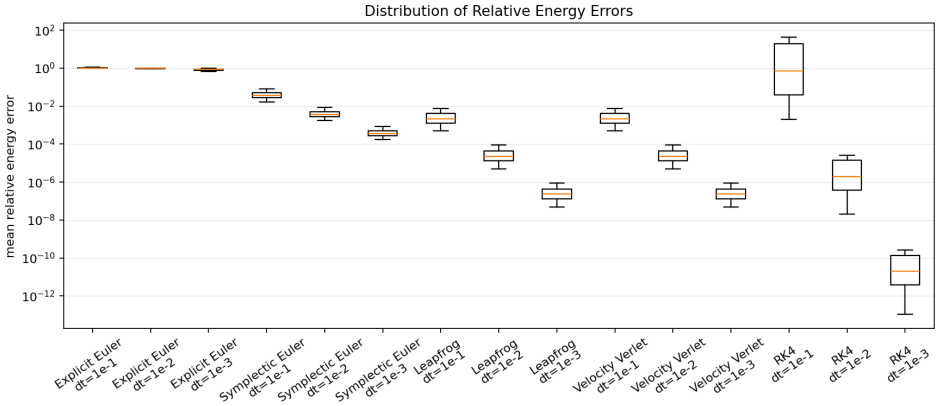

Figure 2: Relative Energy Error Box Plot

across all three timesteps, confirming the boxplot observation that reducing Δt does not meaningfully improve its energy behaviour over the tested range. Symplectic Euler and Velocity Verlet both decrease steadily as the timestep shrinks, with Velocity Verlet consistently outperforming Symplectic Euler by roughly two orders of magnitude at a given resolution. Leapfrog overlaps almost exactly with Velocity Verlet on this plot, which is unsurprising given the two schemes are algebraically equivalent formulations of the same method. RK4 is the standout here: at $\Delta t = 10^{-1}$ it produces the worst energy error of all five integrators, exceeding even Explicit Euler, yet at $\Delta t = 10^{-3}$ it produces the best energy error by several orders of magnitude, dropping below 10^{-8}.

Figures 4 through 6 break the mean, median, and maximum relative energy error down by integrator and resolution as grouped bar charts. All three present essentially the same ranking: at $\Delta t = 10^{-3}$, the ordering from worst to best is Explicit Euler, Symplectic Euler, then Leapfrog and Velocity Verlet together, with RK4 producing the smallest error of the group by a wide margin. The maximum relative energy error (Figure 6) makes the instability of RK4 at coarse timesteps most visible, with a value on the order of 10^4 at $\Delta t = 10^{-1}$, several orders of magnitude above every other integrator at the same resolution.

Finally, Figure 7 shows the mean absolute energy conservation value for each integrator. Explicit Euler, Symplectic Euler, Leapfrog, and Velocity Verlet all sit near zero on this linear scale at every resolution tested. RK4 is the only integrator to depart visibly from zero, and only at $\Delta t = 10^{-1}$, where the mean energy value rises to approximately 0.029, consistent with the instability already noted in Figures 3 and 6.

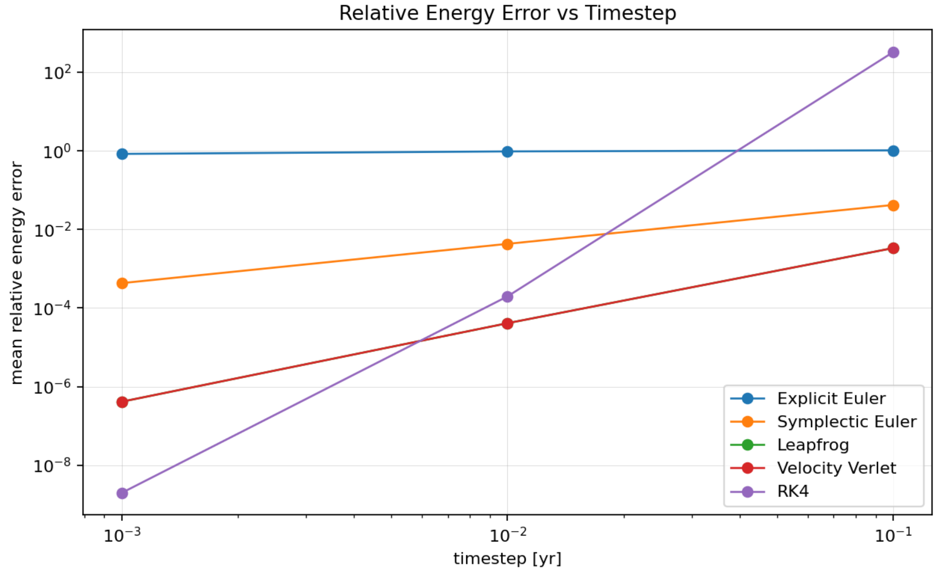

Figure 3: Relative Energy Error vs Timestep

5.3 Angular Momentum Conservation

Figure 8 plots the mean relative angular momentum error against timestep for each integrator. Explicit Euler again remains flat at approximately 10^0 across the tested range, mirroring its behaviour for energy conservation. Symplectic Euler, Leapfrog, and Velocity Verlet all sit near the floating-point noise floor, around 10^{-13}, and are largely insensitive to the choice of timestep, which is consistent with the fact that these are symplectic integrators applied to a central force problem where angular momentum is an exactly conserved quantity of the underlying Hamiltonian. RK4 again shows the steepest dependence on timestep of any integrator, falling from just below 10^0 at $\Delta t = 10^{-1}$ to below 10^{-9} at $\Delta t = 10^{-3}$.

Figures 9 through 11 present the mean, median, and maximum relative angular momentum error broken down by integrator and resolution. Across all three, Symplectic Euler, Leapfrog, and Velocity Verlet remain several orders of magnitude below Explicit Euler and RK4 at every resolution, with their error values hovering around 10^{-13} regardless of timestep. RK4 is the only integrator among the five whose angular momentum error responds strongly to a decrease in timestep, improving from around 10^{-1} at $\Delta t = 10^{-1}$ to around 10^{-9} at $\Delta t = 10^{-3}$ in the mean case.

Figure 12 shows the mean absolute angular momentum value for each integrator. Symplectic Euler, Leapfrog, and Velocity Verlet produce an identical value across all three resolutions, at approximately 7.7×10^{-5}, indicating exact conservation to within floating-point precision independent

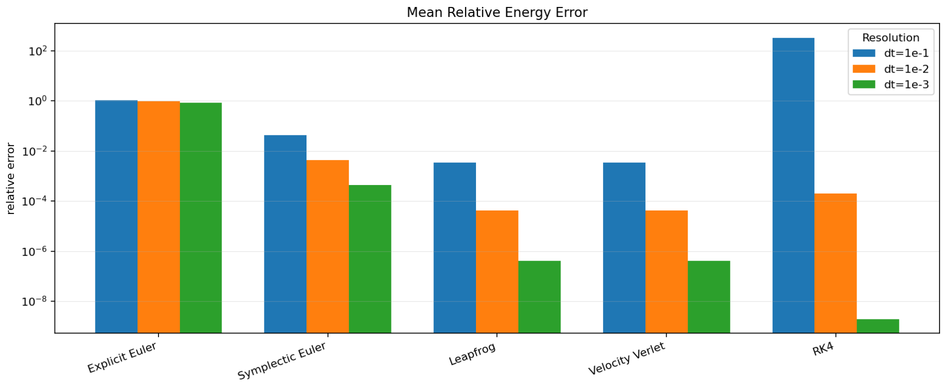

Figure 4: Mean Relative Energy Error by Integrator and Resolution

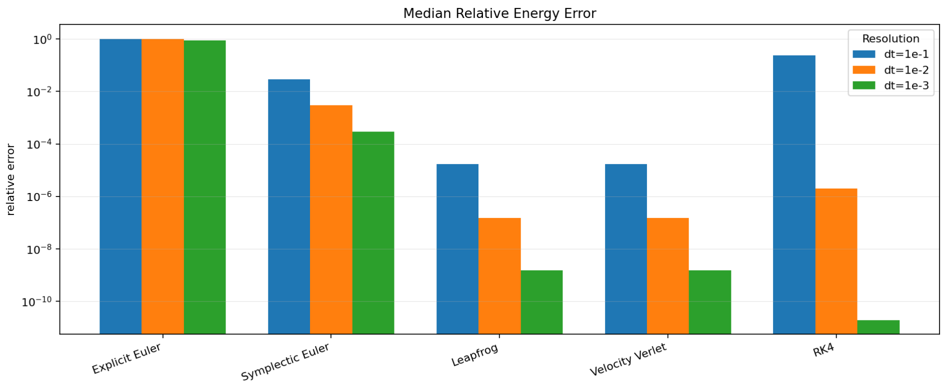

Figure 5: Median Relative Energy Error by Integrator and Resolution

of timestep. Explicit Euler departs from this value and drifts further as the timestep increases, while RK4 also departs from the reference value, most noticeably at $\Delta t = 10^{-1}$.

6 Preliminary Discussion

6.1 Interpretation of the Results

Across all the simulated systems, Explicit Euler consistently showed the poorest conservation of both energy and angular momentum, meanwhile Leapfrog, Velocity Verlet and Symplectic Euler demonstrated substantially greater long-term stability. Although RK4 produced the smallest energy errors at sufficiently small timesteps, its behaviour deteriorated rapidly at coarse resolutions, highlighting the importance of timestep selection for higher-order non-symplectic methods.

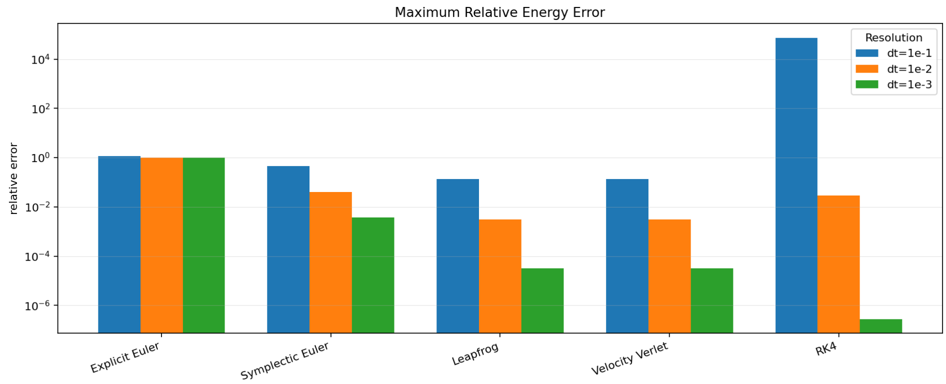

Figure 6: Maximum Relative Energy Error by Integrator and Resolution

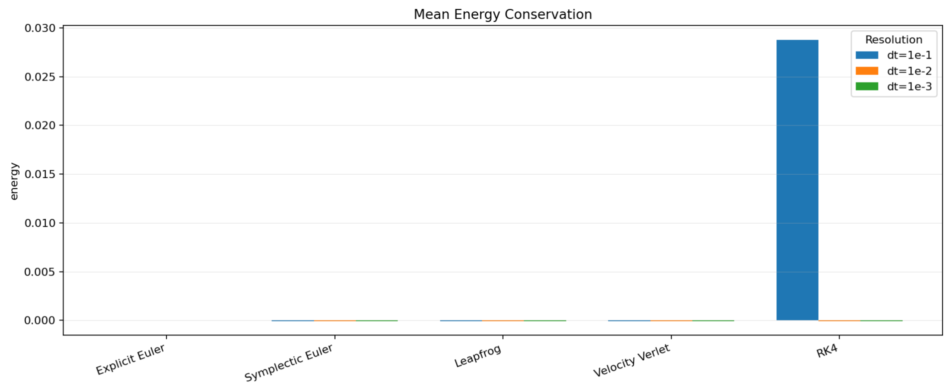

Figure 7: Mean Energy Conservation by Integrator and Resolution

Explicit Euler demonstates such behaviour due to the accumulated truncation errors over each timestep. Symplectic methods by their mathematical design preserve the Hamiltonian geometry. Instead of estimating a precise value of energy, symplectic methods revolve around the true value in the sinusoid. For example, the RK4 algorithm minimises the local truncation error with the higher resolution, yet doesn't preserve the phase space. Hence why, there is a difference between local accuracy and long-term stability.

6.2 Relation to Hamiltonian Mechanics

The observed behaviour closely matches theoretical expectations for Hamiltonian systems. Symplectic methods do not conserve energy exactly; instead they preserve a nearby modified Hamiltonian, resulting in bounded oscillatory error rather than secular drift. This distinction is central to backward error analysis: a symplectic integrator can be shown to solve, up to exponentially small corrections in the timestep, the flow of a shadow Hamiltonian

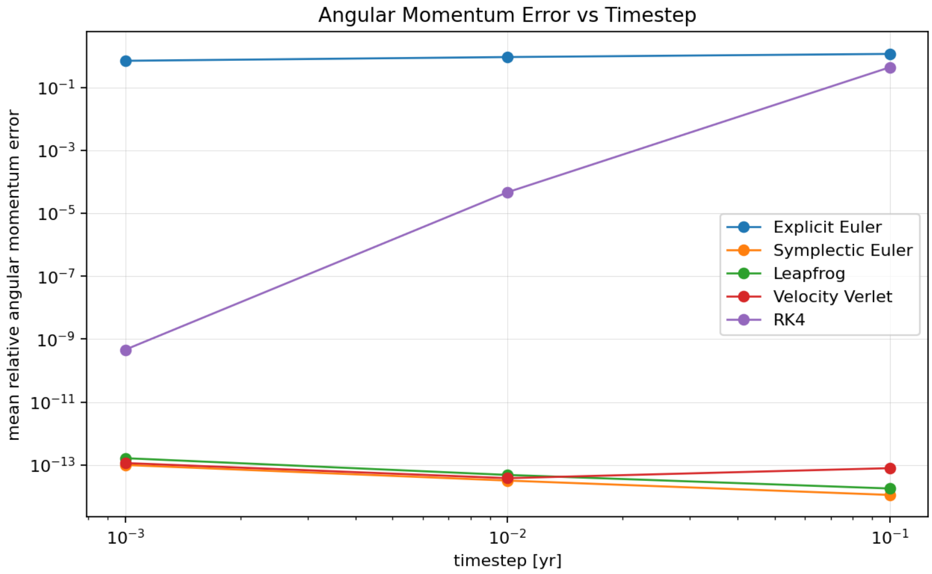

Figure 8: Angular Momentum Error vs Timestep

$\tilde{H}$ that lies close to the true Hamiltonian H. Because $\tilde{H}$ is itself exactly conserved by the discrete map, the numerically computed energy oscillates within a fixed band about the true value rather than drifting away from it. Non-symplectic methods such as RK4 possess no equivalent conserved quantity, so their local truncation errors are free to accumulate coherently over long integrations, producing the secular drift observed at coarse resolutions.

6.3 Angular Momentum Conservation

The conservation of angular momentum provides a complementary, and in some respects more discriminating, test of integrator quality than energy alone. In the continuous two-body problem, angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ is conserved exactly because the gravitational potential depends only on the separation $|\mathbf{r}|$ and is therefore invariant under rotations; by Noether's theorem, this rotational symmetry guarantees a corresponding conserved quantity.

The discrete integrators inherit this symmetry to varying degrees. Symplectic Euler, Leapfrog and Velocity Verlet update velocity using a force evaluated at the current (or half-step) position, and this force is always directed along the instantaneous radius vector. Because the discrete torque $\mathbf{r} \times \mathbf{F}(\mathbf{r})$ vanishes identically at every timestep, exactly as it does in the continuous system, these methods conserve angular momentum to machine precision regardless of the timestep chosen. This is a structural property of the update rule rather than an accuracy effect, and it explains why angular

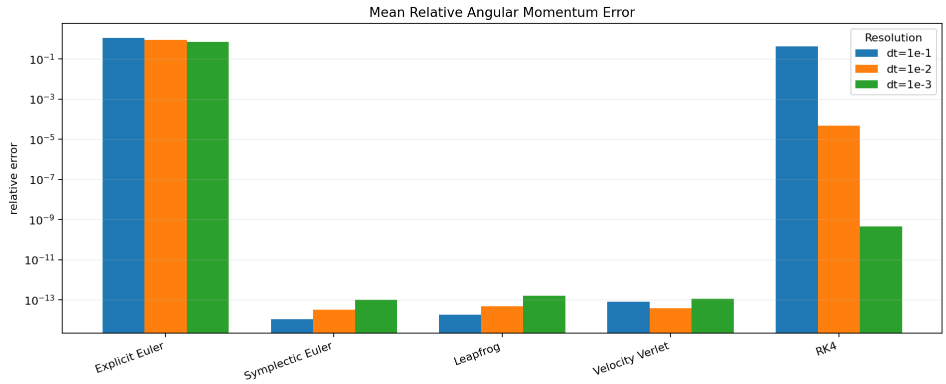

Figure 9: Mean Relative Angular Momentum Error by Integrator and Resolution

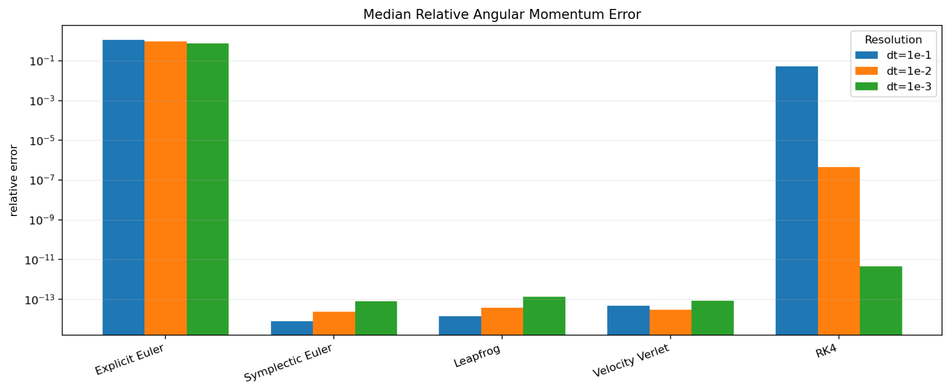

Figure 10: Median Relative Angular Momentum Error by Integrator and Resolution

momentum conservation for the symplectic methods did not degrade even where energy error grew at coarse resolutions.

Explicit Euler and RK4 do not share this property. Explicit Euler updates position and velocity from inconsistent, staggered information (the velocity used to advance position is not the velocity implied by the force just computed), which introduces a small effective torque at each step; this error compounds secularly, matching the drift observed in energy. RK4, despite averaging force evaluations across several sub-stages within a step, does not evaluate the force along a single self-consistent trajectory, so it also fails to preserve $\mathbf{r} \times \mathbf{F} = \mathbf{0}$ exactly at the discrete level. Its angular momentum error was nonetheless substantially smaller than that of Explicit Euler at fine timesteps, consistent with its higher local order of accuracy, but the error was not bounded in the same structural sense as the symplectic methods and would be expected to grow if the timestep were coarsened

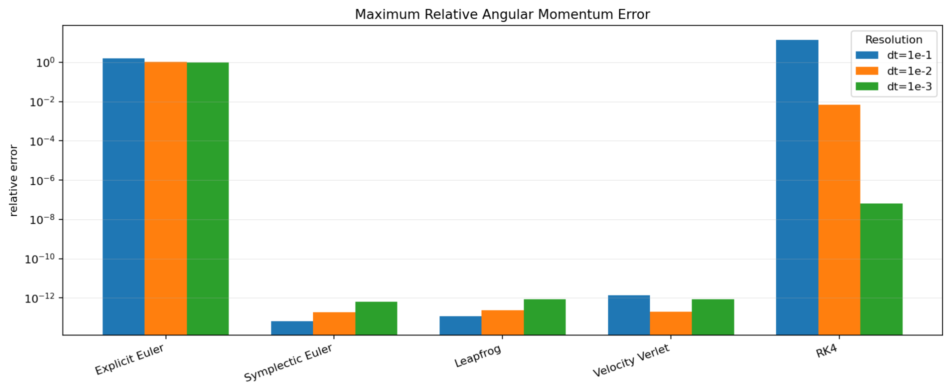

Figure 11: Maximum Relative Angular Momentum Error by Integrator and Resolution

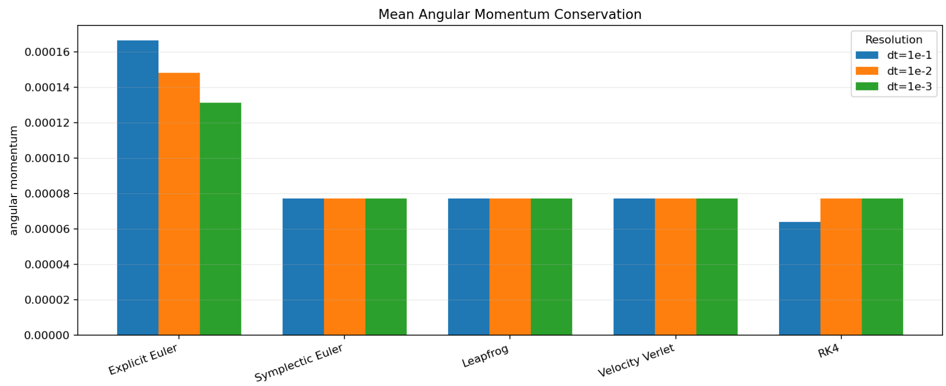

Figure 12: Mean Angular Momentum Conservation by Integrator and Resolution

further or the integration extended over longer timescales. Taken together, the angular momentum results reinforce the central conclusion of this study: symplectic structure, not local order of accuracy, is the primary determinant of long-term qualitative fidelity in Hamiltonian simulations.

6.4 RK4 Behaviour

RK4 possesses fourth-order local accuracy, but this accuracy assumes sufficiently small timesteps. At coarse resolutions, the large integration interval dominates truncation error and the method rapidly loses stability. Once the timestep becomes sufficiently small, the fourth-order convergence dominates, allowing RK4 to outperform all remaining methods in terms of instantaneous energy error.

However, despite this excellent short-term accuracy, RK4 remains non-symplectic and therefore cannot guarantee long-term conservation of invariants in Hamiltonian systems.

6.5 Computational Cost v. Accuracy

Although RK4 produced the lowest numerical error at fine resolution, this came at approximately four force evaluations per timestep and the highest runtime among all methods. Leapfrog and Velocity Verlet therefore provide a considerably better compromise between computational efficiency and conservation accuracy.

6.6 Velocity Verlet v. Leapfrog

Velocity Verlet and Leapfrog are algebraically equivalent formulations of the same second-order symplectic method, differing only in how the half-step velocity is stored and reported, and the results confirm this equivalence at every level examined. In runtime, Leapfrog was marginally the fastest integrator at higher resolutions, with Velocity Verlet close behind it; the two are not expected to differ by more than implementation overhead, since both require one force evaluation per timestep. In energy conservation, Leapfrog was found to overlap almost exactly with Velocity Verlet on the error-versus-timestep comparison, with both methods consistently outperforming Symplectic Euler by roughly two orders of magnitude at a given resolution. The angular momentum results show the same equivalence in its clearest form: both integrators produced an identical mean absolute angular momentum value across all three tested resolutions, and their relative angular momentum error sat at the floating-point noise floor of approximately 10^{-13}, independent of timestep. This lends strong empirical support to the theoretical expectation that the two schemes trace out the same discrete trajectory, and any residual difference between them observed here is attributable to implementation detail (such as the point in the step at which velocity is reported) rather than to a genuine difference in the underlying numerical method. Consequently, neither integrator can be said to be superior to the other on the criteria examined in this study; the choice between them in practice would come down to convenience, for instance whether a synchronised position-velocity pair at each full step (as reported by Velocity Verlet) is more useful downstream than the staggered representation used internally by Leapfrog.

6.7 Hypothesis

The hypothesis was only partially supported. Velocity Verlet demonstrated an excellent balance between runtime and conservation accuracy, closely matching Leapfrog due to their mathematical equivalence. However, RK4 produced lower energy errors at sufficiently fine timesteps, although at substantially greater computational cost and without preserving the symplectic structure.

6.8 Limitations

While the results presented here are consistent with established theory on symplectic integration, several limitations constrain the scope of the conclusions and point towards natural extensions of this work.

Fixed timestep integration. All integrators were evaluated at a single, fixed timestep rather than with an adaptive scheme. A fixed Δt is necessarily a compromise: it must be small enough to resolve the fastest dynamics of the orbit, typically near perihelion where the angular velocity is greatest, while remaining computationally tractable over the full integration window. An adaptive-timestep approach, whether based on local truncation error estimation (as in embedded Runge–Kutta pairs) or on a time-regularising coordinate transformation such as the Sundman transformation, would allow the timestep to shrink automatically in regions of high curvature and expand elsewhere, and would likely narrow the performance gap observed between RK4 and the symplectic methods at coarse resolutions. Adapting a symplectic method to variable timesteps is non-trivial, however, since naive step-size control generally destroys the symplectic property of the map, and this trade-off was not explored here.

Higher-order symplectic compositions. The symplectic methods considered in this study (Symplectic Euler, Leapfrog and Velocity Verlet) are all first- or second-order accurate. Higher-order symplectic integrators can be constructed by composing second-order symplectic maps with carefully chosen step-size coefficients, as in the fourth-order schemes of Forest and Ruth or Yoshida. Such methods would combine the long-term structural stability demonstrated here with markedly reduced local error, and a comparison against them would clarify whether the accuracy advantage of RK4 at fine timesteps persists against a symplectic method of matching order.

Restriction to a two-body system. The analysis was restricted to an isolated two-body Hamiltonian system, for which an analytical Keplerian solution exists and can be used as a ground truth. Extending the comparison to N-body simulations, where no closed-form solution is generally available, would test whether the relative ranking of integrators established here persists in the presence of additional gravitational interactions, and would allow investigation of numerical performance in genuinely chaotic orbital configurations, where nearby trajectories diverge exponentially and long-term structural conservation becomes even more important than local accuracy.

Related settings of particular interest include planetary systems exhibiting mean-motion resonances, where small integration errors can be resonantly amplified over long timescales, and the long-term stability analysis of exoplanetary systems, which typically requires integration over timescales of 10^6–10^9 orbits and is correspondingly sensitive to the choice of integrator.

Computational scaling. The present implementation was run in serial on three cores of a consumer device, and no attempt was made to characterise performance at scale. N-body extensions of this work would benefit from GPU acceleration, which parallelises the pairwise force calculations that dominate the computational cost of large simulations, and from shared- or distributed-memory parallelisation using OpenMP or MPI respectively. A systematic scaling study would be needed to establish whether the computational-cost conclusions drawn here for a single two-body pair generalise to the substantially larger force-evaluation counts required by N-body and resonance studies.

Newtonian approximation. All simulations assumed Newtonian gravity, neglecting relativistic corrections. For systems with strong or close-orbit gravitational fields, such as tight binary star systems or bodies passing close to a compact object, post-Newtonian corrections introduce additional effects, including apsidal precession, that a purely Newtonian integrator cannot reproduce regardless of its symplectic properties. Incorporating these corrections would require modifying the force law itself rather than the integration scheme, and represents a natural extension once the present comparison is established on the simpler Newtonian case.